



Liouville theorems for nonnegative solutions to Hardy–Hénon type system on a half space

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Received: 16 May 2021 / Accepted: 16 November 2021 / Published online: 6 December 2021
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Abstract

In this paper, we are concerned with the Hardy–Hénon type system on a half space $\mathbb{R}_+^n$

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} u(x) = f(x, v), & u(x) \geq 0, & x \in \mathbb{R}_+^n, \\ (-\Delta)^{\frac{\beta}{2}} v(x) = g(x, u), & v(x) \geq 0, & x \in \mathbb{R}_+^n \end{cases}$$

with Dirichlet boundary conditions, where $n \geq 1$, $n > \max\{\alpha, \beta\}$ and $0 < \alpha, \beta \leq 2$. We derive Liouville theorems (i.e., the non-existence of nontrivial nonnegative solutions) provided that f and g satisfy certain subcritical growth conditions (see Theorem 1.6). The argument used in our proof is the method of scaling spheres developed in Dai and Qin (Liouville type theorems for fractional and higher order Hénon–Hardy type equations via the method of scaling spheres, preprint, submitted for publication, p 52, [arXiv: 1810.02752](https://arxiv.org/abs/1810.02752)). Our results generalize the Liouville theorems for single Lane–Emden equation in Chen et al. (Adv Math 274: 167–198, 2015) and Hardy–Hénon type equation in (Dai and Qin [arXiv: 1810.02752](https://arxiv.org/abs/1810.02752)) to system and the Liouville theorems for system of Lane–Emden equations ($f = v^p$, $g = u^q$) in Dai et al. LPotential Anal 46:569–588, 2017) to system of equations with general Hardy–Hénon type nonlinearities $f(x, v)$ and $g(x, u)$.

Keywords Fractional Laplacians · Hardy–Hénon type system · Nonnegative solutions · Liouville theorems · Method of scaling spheres

Mathematics Subject Classification 35B53 · 35J30 · 35J91

Communicated by Julian Bonder.

Wei Dai is supported by the NNSF of China (No. 11971049), the Fundamental Research Funds for the Central Universities, and the State Scholarship Fund of China (No. 201806025011).

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1 Introduction

In this paper, we will investigate the Liouville property of nonnegative solutions to Hénon–Hardy type system with Dirichlet boundary conditions on a half space $\mathbb{R}_+^n$

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} u(x) = f(x, v), & u(x) \geq 0, & x \in \mathbb{R}_+^n, \\ (-\Delta)^{\frac{\beta}{2}} v(x) = g(x, u), & v(x) \geq 0, & x \in \mathbb{R}_+^n, \\ u(x) = v(x) = 0, & & x \in \mathbb{R}^n \setminus \mathbb{R}_+^n, \end{cases} \quad (1.1)$$

where $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, $\mathbb{R}_+^n = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n > 0\}$ is the upper half Euclidean space, the nonlinear terms $f : \mathbb{R}_+^n \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ and $g : \mathbb{R}_+^n \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$.

When $0 < \alpha < 2$, the nonlocal fractional Laplacians $(-\Delta)^{\frac{\alpha}{2}}$ are defined by

$$(-\Delta)^{\frac{\alpha}{2}} u(x) = C_{\alpha,n} P.V. \int_{\mathbb{R}^n} \frac{u(x) - u(y)}{|x - y|^{n+\alpha}} dy := C_{\alpha,n} \lim_{\epsilon \rightarrow 0} \int_{|y-x| \geq \epsilon} \frac{u(x) - u(y)}{|x - y|^{n+\alpha}} dy, \quad (1.2)$$

for any functions $u \in C_{loc}^{1,1} \cap \mathcal{L}_\alpha(\mathbb{R}^n)$, where $C_{\alpha,n} = \left(\int_{\mathbb{R}^n} \frac{1 - \cos(2\pi \zeta_1)}{|\zeta|^{n+\alpha}} d\zeta \right)^{-1}$ and the function spaces

$$\mathcal{L}_\alpha(\mathbb{R}^n) := \left\{ u : \mathbb{R}^n \rightarrow \mathbb{R} \mid \int_{\mathbb{R}^n} \frac{|u(x)|}{1 + |x|^{n+\alpha}} dx < \infty \right\}. \quad (1.3)$$

The fractional Laplacians $(-\Delta)^{\frac{\alpha}{2}}$ can also be defined equivalently (see [10]) by Caffarelli and Silvestre's extension method (see [15]) for $u \in C_{loc}^{1,1}(\mathbb{R}_+^n) \cap \mathcal{L}_\alpha(\mathbb{R}^n)$.

We assume the classical solution $u \in \mathcal{L}_\alpha(\mathbb{R}^n) \cap C_{loc}^{1,1}(\mathbb{R}_+^n) \cap C(\overline{\mathbb{R}_+^n})$ if $0 < \alpha < 2$ and $u \in C^2(\mathbb{R}_+^n) \cap C(\overline{\mathbb{R}_+^n})$ if $\alpha = 2$, $v \in \mathcal{L}_\beta(\mathbb{R}^n) \cap C_{loc}^{1,1}(\mathbb{R}_+^n) \cap C(\overline{\mathbb{R}_+^n})$ if $0 < \beta < 2$ and $v \in C^2(\mathbb{R}_+^n) \cap C(\overline{\mathbb{R}_+^n})$ if $\beta = 2$. For any $0 < \alpha < 2$, the function spaces $\mathcal{L}_\alpha(\mathbb{R}^n)$ are defined by

$$\mathcal{L}_\alpha(\mathbb{R}^n) := \left\{ u : \mathbb{R}^n \rightarrow \mathbb{R} \mid \int_{\mathbb{R}^n} \frac{|u(x)|}{1 + |x|^{n+\alpha}} dx < +\infty \right\}. \quad (1.4)$$

The fractional Laplacian is a nonlocal integral-differential operator. It can be used to model various physical phenomena, such as turbulence and water waves, relativistic quantum mechanics of stars, molecular dynamics, and anomalous diffusion and quasi-geostrophic flows (see [14, 17] and the references therein). The fractional Laplacian also has various applications in probability and finance (see [1, 16] and the references therein). In particular, it can also be understood as the infinitesimal generator of a stable Lévy process (see [1]).

To introduce assumptions on the nonlinearities $f(x, v)$ and $g(x, u)$, we need the following definitions.

Definition 1.1 We say that the nonlinear term f has subcritical (critical) growth provided that

$$\mu^{\frac{n+\alpha}{n-\alpha}} f\left(\mu^{\frac{2}{n-\alpha}} x, \mu^{-\frac{n-\beta}{n-\alpha}} v\right) \quad (1.5)$$

is strictly increasing (invariant) with respect to $\mu \geq 1$ or $\mu \leq 1$ for all $(x, v) \in \mathbb{R}_+^n \times \mathbb{R}_+$. The corresponding definitions for $g(x, u)$ can be given similarly by exchanging α and β .

Definition 1.2 A function $h(x, u)$ is called locally Lipschitz on u in $\mathbb{R}_+^n \times \overline{\mathbb{R}_+}$, provided that for any $u_0 \in \overline{\mathbb{R}_+}$ and $\Omega \subseteq \mathbb{R}_+^n$ bounded, there exists a (relatively) open neighborhood $U(u_0) \subset \overline{\mathbb{R}_+}$, such that h is Lipschitz continuous on u in $\Omega \times U(u_0)$.

Throughout this paper, we assume that the nonlinear terms $f(x, v)$ and $g(x, u)$ have subcritical or critical growth in the sense of Definition 1.1.

We need the following three assumptions on the nonlinear terms $f(x, v)$ and $g(x, u)$.

(A₁) If $f(x, u)$ has subcritical growth, then it is nondecreasing about u in $\mathbb{R}_+^n \times \overline{\mathbb{R}_+}$, namely

$$(x, u), (x, v) \in \mathbb{R}_+^n \times \overline{\mathbb{R}_+} \text{ with } u \leq v \text{ implies } f(x, u) \leq f(x, v). \quad (1.6)$$

If $f(x, u)$ has critical growth, then it is strictly increasing about u in $\mathbb{R}_+^n \times \overline{\mathbb{R}_+}$, namely,

$$(x, u), (x, v) \in \mathbb{R}_+^n \times \overline{\mathbb{R}_+} \text{ with } u < v \text{ implies } f(x, u) < f(x, v). \quad (1.7)$$

The same assumptions are also made on the nonlinear term $g(x, u)$.

(A₂) There exist $\sigma < \alpha$ and $\theta < \beta$, such that $|x|^\sigma f(x, u)$ and $|x|^\theta g(x, u)$ are locally Lipschitz on u in $\mathbb{R}_+^n \times \overline{\mathbb{R}_+}$.

(A₃) There exist a cone $\mathcal{C} \subset \Omega$ with vertex at 0 which contains the positive x_n -axis (if $n = 1$, $\mathcal{C} = \mathbb{R}_+$; if $n = 2$, for instance, $\mathcal{C} = \left\{x \in \mathbb{R}_+^n \mid x_n > \frac{|x|}{\sqrt{n}}\right\}$), constants $C > 0$, $-\alpha < a < +\infty$, $-\beta < b < +\infty$, $1 \leq p \leq p_c(a) := \frac{n+\alpha+2a}{n-\beta}$ and $1 \leq q \leq q_c(b) := \frac{n+\beta+2b}{n-\alpha}$, such that the nonlinear term

$$f(x, v) \geq \overline{C}|x|^a v^p, \quad g(x, u) \geq \overline{C}|x|^b u^q \quad (1.8)$$

in $\mathcal{C} \times \overline{\mathbb{R}_+}$.

In particular, if we set $\alpha = \beta$ and $f = g$, PDEs' system (1.1) will degenerate into one single Hardy–Hénon type equation

$$(-\Delta)^{\frac{a}{2}} u(x) = f(x, u), \quad x \in \mathbb{R}_+^n. \quad (1.9)$$

If $f(x, u) = |x|^a u^p$, equations (1.9) are called Hénon, Lane–Emden, Hardy equations for $a > 0$, $a = 0$, $a < 0$, respectively. These equations have diverse and important applications in conformal geometry and Sobolev inequalities. In particular, in the case $a = 0$, (1.9) becomes the well-known Lane–Emden equation, which models many phenomena in mathematical physics and astrophysics.

Liouville type theorems for Eq. (1.9) (i.e., the non-existence of nontrivial non-negative solutions) in the whole space $\mathbb{R}^n$ and the half space $\mathbb{R}_+^n$ have been

extensively studied (c.f. [3–5, 9, 12, 13, 19, 20, 22–26, 29, 30, 33, 34, 36–38, 41, 42, 48, 49] and the references therein). These Liouville theorems, in conjunction with the blowing up and re-scaling arguments, are crucial in establishing a priori estimates and hence existence of positive solutions to non-variational boundary value problems for a class of elliptic equations on bounded domains or Riemannian manifolds with boundaries (c.f. [23, 32, 40]). In [16], Cabré and Tan deal with equations involving a different kind of fractional operator $A_{\frac{1}{2}}$. The fractional operators A_s ($s \in (0, 1)$) are defined using the spectral decomposition of the Laplacian $-\Delta$, i.e., $A_s u := \sum_{k=1}^{\infty} a_k \lambda_k^s e_k$, where λ_k and e_k are eigenvalues and the corresponding eigenfunctions for the Laplacian operator $-\Delta$ in Ω with homogeneous Dirichlet boundary data, while a_k represents the projection of u on the direction e_k . Servadei and Valdinoci compares the fractional Laplacians $(-\Delta)^s$ and the fractional operators A_s especially from the spectrum point of view and give their differences in [44]. The first analysis of fractional systems was carried out by Dipierro and Pinamonti in [21]. For more works on the Liouville type theorems for system of Hardy-Hénon type equations, see, e.g., [2, 19, 28, 37, 40, 46, 47, 49] and the references therein.

In this paper, using the method of scaling spheres developed by Dai and Qin in [23] (see also [4, 24–26]), we prove Liouville type theorems for nonnegative solutions to the Hardy-Hénon type system (1.1) provided that the nonlinear terms f and g satisfy the subcritical conditions (in the sense of Definition 1.1) and assumptions (A_1) , (A_2) , and (A_3) .

First, we need to show that the nonnegative classical solution (u, v) to PDEs' system (1.1) also satisfies the following equivalent IEs' system:

$$\begin{cases} u(x) = \int_{\mathbb{R}_+^n} G_{\alpha}^{+}(x, y) f(y, v(y)) dy, \\ v(x) = \int_{\mathbb{R}_+^n} G_{\beta}^{+}(x, y) g(y, u(y)) dy, \end{cases} \quad (1.10)$$

where $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, and the Green function associated with Dirichlet problem for $(-\Delta)^{\frac{\gamma}{2}}$ ($\gamma \in (0, 2] \cap (0, n)$) on $\mathbb{R}_+^n$ is given by

$$G_{\gamma}^{+}(x, y) := \frac{C_{n, \gamma}}{|x - y|^{n-\gamma}} \int_0^{\frac{4x_n y_n}{|x - y|^2}} \frac{h^{\frac{\gamma}{2}-1}}{(1+h)^{\frac{n}{2}}} dh. \quad (1.11)$$

We have the following equivalence between PDEs' system (1.1) and IEs' system (1.10).

Theorem 1.3 *Assume $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, $f(x, v)$ and $g(x, u)$ satisfy the assumption (A_3) . Suppose (u, v) is a pair of nonnegative solution to PDEs' system (1.1), then (u, v) also solves the IEs' system (1.10), and vice versa.*

Remark 1.4 *From the proof of Theorem 1.3 (see Sect. 2), one can observe that, if $f(x, v)$ and $g(x, u)$ satisfy the assumption (A_3) with general $1 \leq p < +\infty$ and $1 \leq q < +\infty$, Theorem 1.3 is still valid.*

Next, by applying the method of scaling spheres developed in [23] (see also, e.g., [4, 23–26]), we can establish the following Liouville type result for IEs' system (1.10). For more literature on the classification of solutions and Liouville type theorems for various PDE and IE problems via the methods of moving planes or spheres, please refer to [5, 6, 8, 11, 12, 18, 20, 22, 31, 33–36, 39, 43, 48, 49] and the references therein.

Theorem 1.5 Assume $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, $f(x, v)$ and $g(x, u)$ satisfy assumptions (A_1) , (A_2) and (A_3) . Suppose $(u, v) \in C(\mathbb{R}_+^n) \times C(\mathbb{R}_+^n)$ is a pair of nonnegative solution to IEs' system (1.10) on $\mathbb{R}_+^n$, if $f(x, v)$ is subcritical and satisfies (A_3) with $1 \leq p < \frac{n+\alpha+2a}{n-\beta}$ or $g(x, u)$ is subcritical and satisfies (A_3) with $1 \leq q < \frac{n+\beta+2b}{n-\alpha}$, then $(u, v) \equiv (0, 0)$ in $\mathbb{R}_+^n$.

As a consequence of the equivalence between PDEs' system (1.1) and IEs' system (1.10) in Theorem 1.3 and the Liouville type result for IEs' system in Theorem 1.5, we derive immediately the following Liouville theorem for PDEs' system (1.1).

Theorem 1.6 Assume $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, $f(x, v)$ and $g(x, u)$ satisfy assumptions (A_1) , (A_2) and (A_3) . Suppose (u, v) is a pair of nonnegative classical solution to PDEs' system (1.1) on $\mathbb{R}_+^n$, if $f(x, v)$ is subcritical and satisfies (A_3) with $1 \leq p < \frac{n+\alpha+2a}{n-\beta}$ or $g(x, u)$ is subcritical and satisfies (A_3) with $1 \leq q < \frac{n+\beta+2b}{n-\alpha}$, then $(u, v) \equiv (0, 0)$ in $\mathbb{R}_+^n$.

Remark 1.7 Theorem 1.6 generalizes the Liouville theorems for single Hardy–Hénon type equation on $\mathbb{R}_+^n$ in [23] to system and the Liouville theorems for system of Lane–Emden equations ($f = v^p$, $g = u^q$) on $\mathbb{R}_+^n$ in [19] to system of equations with general Hardy–Hénon type nonlinearities $f(x, v)$ and $g(x, u)$.

In particular, $f(x, v) = |x|^{a_1}(x_n)^{a_2}v^p$ and $g(x, u) = |x|^{b_1}(x_n)^{b_2}u^q$ with $-\alpha < a_1 + a_2 < +\infty$, $-\beta < b_1 + b_2 < +\infty$, $a_2, b_2 \geq 0$, $1 \leq p \leq \frac{n+\alpha+2(a_1+a_2)}{n-\beta}$ and $1 \leq q \leq \frac{n+\beta+2(b_1+b_2)}{n-\alpha}$ satisfy all the assumptions in Theorems 1.3, 1.5 and 1.6. Thus, we can deduce the following Corollary from Theorem 1.6.

Corollary 1.8 Assume $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$. Suppose that, in PDEs' system (1.1), the nonlinearities $f(x, v) = |x|^{a_1}(x_n)^{a_2}v^p$ and $g(x, u) = |x|^{b_1}(x_n)^{b_2}u^q$, where $-\alpha < a_1 + a_2 < +\infty$, $-\beta < b_1 + b_2 < +\infty$, $a_2, b_2 \geq 0$, $1 \leq p \leq \frac{n+\alpha+2(a_1+a_2)}{n-\beta}$, $1 \leq q \leq \frac{n+\beta+2(b_1+b_2)}{n-\alpha}$ and $(p, q) \neq \left(\frac{n+\alpha+2(a_1+a_2)}{n-\beta}, \frac{n+\beta+2(b_1+b_2)}{n-\alpha}\right)$. Then, if (u, v) is a pair of nonnegative classical solution to PDEs' system (1.1) on $\mathbb{R}_+^n$, we must have $(u, v) \equiv (0, 0)$ in $\mathbb{R}_+^n$.

Remark 1.9 Although systems involving arbitrary number of equations can also be treated in similar way, we mainly focus on the system consisting of two equations in this paper for the sake of simplicity. To derive the equivalence between the PDEs'

system (1.1) and the IEs' system (1.10), we require that $(-\Delta)^{\frac{\alpha}{2}}u$ and $(-\Delta)^{\frac{\beta}{2}}v$ are defined point-wisely and continuous (see, e.g., (2.4) and Lemma 2.1). Thus, the pair of solutions (u, v) are assumed to be classical in Theorems 1.3 and 1.6. To carry out the method of scaling spheres to derive Liouville theorem for the equivalent IEs' system (1.10), we need to assume the continuity on the pair of solutions (u, v) (see, e.g., (3.1), (3.37), (3.38), and Theorem 3.1).

In the following, we will use C to denote a general positive constant that may depend on $n, \alpha, \beta, \sigma, \theta, a, b, p, q, \bar{C}, u$ and the cone $\mathcal{C}$, and whose value may differ from line to line.

The rest of this paper is arranged as follows. In Section 2, we establish the equivalence between PDEs' system (1.1) and IEs' system (1.10). In Section 3, we will use the method of scaling spheres to prove Theorem 1.5, that is, the Liouville theorem for IEs' system (1.10).

2 The equivalence between PDEs' system and IEs' system

In this section, inspired by the ideas from [5, 30, 49] (see also [19, 23]), we will establish the equivalence between PDEs' system (1.1) and IEs' system (1.10) (Theorem 1.3). Without loss of generality, we may assume the cone $\mathcal{C} = \left\{x \in \mathbb{R}_+^n \mid x_n > \frac{|x|}{\sqrt{n}}\right\}$ if $n \geq 2$ and $\mathcal{C} = \mathbb{R}_+$ if $n = 1$ in assumption (A_3) .

First, assume (u, v) is a pair of nonnegative classical solutions to PDEs' system (1.1), our goal is to show that (u, v) also satisfies IEs' system (1.10). For arbitrary $R > 0$, let $P_R := (0, \dots, 0, R)$ and

$$\begin{cases} \tilde{u}_R(x) = \int_{B_R(P_R)} G_R^\alpha(x, y) f(y, v(y)) dy, \\ \tilde{v}_R(x) = \int_{B_R(P_R)} G_R^\beta(x, y) g(y, u(y)) dy, \end{cases} \quad (2.1)$$

where the Green function G_R^γ for $(-\Delta)^{\frac{\gamma}{2}}$ ($0 < \gamma \leq 2$) on $B_R(P_R)$ is given by (c.f. [5, 10])

$$G_R^\gamma(x, y) := \frac{C_{n, \gamma}}{|x - y|^{n-\gamma}} \int_0^{\frac{t_R}{s_R}} \frac{b^{\frac{\gamma}{2}-1}}{(1+b)^{\frac{n}{2}}} db, \quad (2.2)$$

where $t_R := \left(1 - \frac{|x - P_R|^2}{R^2}\right) \left(1 - \frac{|y - P_R|^2}{R^2}\right)$ and $s_R := \frac{|x - y|^2}{R^2}$.

Then, we can derive

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} \tilde{u}_R(x) = f(x, v(x)), & x \in B_R(P_R), \\ (-\Delta)^{\frac{\beta}{2}} \tilde{v}_R(x) = g(x, u(x)), & x \in B_R(P_R), \\ \tilde{u}_R(x) = \tilde{v}_R(x) = 0, & x \notin B_R(P_R). \end{cases} \quad (2.3)$$

Let $U_R(x) = u(x) - \tilde{u}_R(x)$ and $V_R(x) = v(x) - \tilde{v}_R(x)$, by PDEs' system (1.1); we have

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} U_R(x) = 0, & x \in B_R(P_R), \\ (-\Delta)^{\frac{\beta}{2}} V_R(x) = 0, & x \in B_R(P_R), \\ U_R(x) \geq 0, \quad V_R(x) \geq 0, & x \notin B_R(P_R). \end{cases} \quad (2.4)$$

Now, we need the following maximum principle for fractional Laplacians.

Lemma 2.1 (Maximum principle, [9, 45]) *Let Ω be a bounded domain in $\mathbb{R}^n$ and $0 < \alpha < 2$. Assume that $u \in \mathcal{L}_\alpha \cap C_{\text{loc}}^{1,1}(\Omega)$ and is l.s.c. on $\overline{\Omega}$. If $(-\Delta)^{\frac{\alpha}{2}} u \geq 0$ in Ω and $u \geq 0$ in $\mathbb{R}^n \setminus \Omega$, then $u \geq 0$ in $\mathbb{R}^n$. Moreover, if $u = 0$ at some point in Ω , then $u = 0$ a.e. in $\mathbb{R}^n$. These conclusions also hold for unbounded domain Ω if we assume further that*

$$\liminf_{|x| \rightarrow \infty} u(x) \geq 0.$$

By Lemma 2.1 and maximum principles for $-\Delta$, we deduce from (2.4) that, for any $x \in B_R(P_R)$

$$U_R(x) \geq 0, \quad V_R(x) \geq 0. \quad (2.5)$$

Letting $R \rightarrow \infty$, we have

$$u(x) \geq \int_{\mathbb{R}_+^n} G_\alpha^+(x, y) f(y, v(y)) dy =: \tilde{u}(x), \quad (2.6)$$

$$v(x) \geq \int_{\mathbb{R}_+^n} G_\beta^+(x, y) g(y, u(y)) dy =: \tilde{v}(x). \quad (2.7)$$

One can observe that $(\tilde{u}, \tilde{v})$ is a pair of nonnegative solution to

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} \tilde{u}(x) = f(x, v(x)), & x \in \mathbb{R}_+^n, \\ (-\Delta)^{\frac{\beta}{2}} \tilde{v}(x) = g(x, u(x)), & x \in \mathbb{R}_+^n, \\ \tilde{u}(x) = \tilde{v}(x) = 0, & x \notin \mathbb{R}_+^n. \end{cases} \quad (2.8)$$

Define $U := u - \tilde{u}$ and $V := v - \tilde{v}$, and we have

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} U(x) = 0, \quad U(x) \geq 0, & x \in \mathbb{R}_+^n, \\ (-\Delta)^{\frac{\beta}{2}} V(x) = 0, \quad V(x) \geq 0, & x \in \mathbb{R}_+^n, \\ U(x) = V(x) = 0, & x \notin \mathbb{R}_+^n. \end{cases} \quad (2.9)$$

Now, we need the following Lemma from [13] on Liouville theorem for α -harmonic functions on $\mathbb{R}_+^n$.

Lemma 2.2 ([13]) *Assume $0 < \alpha < 2$, $u \in L_{\text{loc}}^\infty(\overline{\mathbb{R}_+^n}) \cap \mathcal{L}_\alpha(\mathbb{R}^n)$ satisfies the following equation in the sense of distribution:*

$$\begin{cases} (-\Delta)^{\frac{\alpha}{2}} u(x) = 0, & u(x) \geq 0, \quad x \in \mathbb{R}_+^n, \\ u(x) \equiv 0, & x \notin \mathbb{R}_+^n. \end{cases} \quad (2.10)$$

Then, either $u \equiv 0$ or

$$\begin{cases} u(x) = Cx_n^{\frac{\alpha}{2}}, & x \in \mathbb{R}_+^n, \\ u(x) = 0, & x \notin \mathbb{R}_+^n, \end{cases} \quad (2.11)$$

for some positive constant $C > 0$.

The Liouville theorem for harmonic functions on $\mathbb{R}_+^n$ (i.e., Lemma 2.2 with $\alpha = 2$) also holds (see, e.g., [27]).

From Lemma 2.2 and Liouville theorem for harmonic functions on $\mathbb{R}_+^n$, we can deduce that either

$$U(x) = V(x) = 0, \quad \forall x \in \mathbb{R}_+^n, \quad (2.12)$$

or there exist two positive constants C_1 and C_2 , such that

$$U(x) = C_1 x_n^{\frac{\alpha}{2}} \text{ or } V(x) = C_2 x_n^{\frac{\beta}{2}}, \quad \forall x \in \mathbb{R}_+^n. \quad (2.13)$$

Without loss of generality, we may assume $U(x) = C_1 x_n^{\frac{\alpha}{2}}$ in $\mathbb{R}_+^n$ in the second case. We will obtain a contradiction by deriving a lower bound estimate of the Green function G_γ^+ ($0 < \gamma \leq 2$). In fact, for each fixed $x \in \mathbb{R}_+^n$, by (1.11), there exists a $R_x > 0$ sufficiently large, such that, for any $|y| \geq R$, one can derive

$$\begin{aligned} G_\gamma^+(x, y) &= \frac{C_{n,\gamma}}{|x-y|^{n-\gamma}} \int_0^{\frac{4x_n y_n}{|x-y|^2}} \frac{b^{\frac{\gamma}{2}-1}}{(1+b)^{\frac{n}{2}}} db \\ &\geq \frac{C}{|x-y|^{n-\gamma}} \int_0^{\frac{4x_n y_n}{|x-y|^2}} b^{\frac{\gamma}{2}-1} db \\ &\geq \frac{C}{|x-y|^{n-\gamma}} \left(\frac{x_n y_n}{|x-y|^2} \right)^{\frac{\gamma}{2}} \\ &\geq \frac{C(x_n y_n)^{\frac{\gamma}{2}}}{|x-y|^n}. \end{aligned} \quad (2.14)$$

Therefore, for each fixed $x = (0, \dots, 0, x_n) \in \mathbb{R}_+^n$, we get from (2.14) and the assumption (A_3) on $f(x, v)$ and $g(x, u)$ that

$$\begin{aligned}
 v(x) &\geq \tilde{v}(x) = \int_{\mathbb{R}_+^n} G_\beta^+(x, y) g(y, u(y)) dy \\
 &\geq \bar{C} C_1^q \int_{\mathbb{R}_+^n \cap \mathcal{C}} G_\beta^+(x, y) |y|^b \left(y_n^{\frac{\beta}{2}} \right)^q dy \\
 &\geq C x_n^{\frac{\beta}{2}} \int_{(\mathbb{R}_+^n \setminus B_R(0)) \cap \mathcal{C}} \frac{y_n^{\frac{\beta(q+1)}{2}} |y|^b}{|x - y|^n} dy \\
 &\geq C x_n^{\frac{\beta}{2}} \int_R^\infty y_n^{\frac{\beta(q+1)}{2}} \int_0^{\sqrt{n-1} y_n} \frac{r^{n-2} (r^2 + y_n^2)^{\frac{b}{2}}}{(r^2 + |x_n - y_n|^2)^{\frac{n}{2}}} dr dy_n \\
 &\geq C x_n^{\frac{\beta}{2}} \int_R^\infty y_n^{\frac{\beta(q+1)}{2} + b - 1} \int_0^{\sqrt{n-1}} \frac{s^{n-2}}{(s^2 + 1)^{\frac{n-b}{2}}} ds dy_n = +\infty,
 \end{aligned} \tag{2.15}$$

which is absurd. This implies that the second case (2.13) can not happen. Therefore, we can derive from (2.12) that

$$\begin{cases} u(x) = \tilde{u}(x) = \int_{\mathbb{R}_+^n} G_\alpha^+(x, y) f(y, v(y)) dy, \\ v(x) = \tilde{v}(x) = \int_{\mathbb{R}_+^n} G_\beta^+(x, y) g(y, u(y)) dy, \end{cases} \tag{2.16}$$

that is, (u, v) also satisfies the IEs' system (1.10).

Conversely, assume that (u, v) is a pair of classical solution to IEs' system (1.10) on $\mathbb{R}_+^n$, and then

$$\begin{aligned}
 (-\Delta)^{\frac{\alpha}{2}} u(x) &= \int_{\mathbb{R}_+^n} \left[(-\Delta)^{\frac{\alpha}{2}} G_\alpha^+(x, y) \right] f(y, v(y)) dy \\
 &= \int_{\mathbb{R}_+^n} \delta(x - y) f(y, v(y)) dy = f(x, v(x)), \quad \forall x \in \mathbb{R}_+^n,
 \end{aligned} \tag{2.17}$$

$$\begin{aligned}
 (-\Delta)^{\frac{\beta}{2}} v(x) &= \int_{\mathbb{R}_+^n} \left[(-\Delta)^{\frac{\beta}{2}} G_\beta^+(x, y) \right] g(y, u(y)) dy \\
 &= \int_{\mathbb{R}_+^n} \delta(x - y) g(y, u(y)) dy = g(x, u(x)), \quad \forall x \in \mathbb{R}_+^n.
 \end{aligned} \tag{2.18}$$

Therefore, (u, v) also solves the PDEs' system (1.1). This completes the proof of Theorem 1.3.

3 Proof of Theorem 1.5

In this section, we will prove Theorem 1.5 via contradiction arguments and the method of scaling spheres. Without loss of generality, we may assume that $f(x, v)$ is subcritical and satisfies (A_3) with $1 \leq p < \frac{n+\alpha+2a}{n-\beta}$, the cone $\mathcal{C} = \left\{ x \in \mathbb{R}_+^n \mid x_n > \frac{|x|}{\sqrt{n}} \right\}$ if $n \geq 2$ and

$C = \mathbb{R}_+$ if $n = 1$ in assumption (A_3) . We may also assume that the nonlinear terms $f(x, v)$ and $g(x, u)$ satisfy subcritical or critical conditions in Definition 1.1 for $\mu \geq 1$. If $f(x, v)$ and $g(x, u)$ satisfy subcritical or critical conditions in Definition 1.1 for $\mu \leq 1$, we only need to carry out calculations and estimates outside the upper half ball $B_\lambda(0) \cap \mathbb{R}_+^n$ during the scaling spheres procedure.

The method of scaling spheres mainly consists of three steps. First, the IEs' system (1.10) implies that positive solution (u, v) satisfies the same lower bound as fundamental solutions as $|x| \rightarrow +\infty$. Second, by combining this lower bound with the IEs' system (1.10) and the scaling spheres procedure, we will get a better lower bound for positive solution than fundamental solutions. Third, based on the second step, a "Bootstrap" iteration process in conjunction with the IEs' system (1.10) will provide better and better lower bound estimates on the asymptotic behavior of positive solution (u, v) (see Theorem 3.1), which finally lead to a contradiction with the integrability of solutions indicated by the IEs' system (1.10) unless the solution $(u, v) \equiv (0, 0)$.

Now, suppose on the contrary that (u, v) is a pair of nonnegative solution to the equivalent IEs' system (1.10), but (u, v) is not identically zero, then one can infer from the IEs' system (1.10) that (u, v) is actually a pair of positive solution, i.e., $u > 0$ and $v > 0$ in $\mathbb{R}_+^n$. Moreover, there exists a constant $C > 0$, such that the positive solution (u, v) satisfies the following lower bound:

$$u(x) \geq C \frac{x_n^{\frac{\alpha}{2}}}{|x|^n}, \quad v(x) \geq C \frac{x_n^{\frac{\alpha}{2}}}{|x|^n}, \quad \forall |x| \geq 1, \quad x \in \mathbb{R}_+^n. \quad (3.1)$$

Indeed, since $u > 0$ and $v > 0$ satisfy the IEs' system (1.10), we can deduce from assumption (A_3) that

$$\begin{aligned} u(x) &= \int_{\mathbb{R}_+^n} \frac{C_{n,\alpha}}{|x-y|^{n-\alpha}} \left(\int_0^{\frac{4x_n y_n}{|x-y|^2}} \frac{h^{\frac{\alpha}{2}-1}}{(1+h)^{\frac{n}{2}}} dh \right) f(y, v(y)) dy \\ &\geq C \int_{\mathbb{R}_+^n \cap \{|y| \leq \frac{1}{2}\}} \frac{1}{|x-y|^{n-\alpha}} \left(\int_0^{\frac{4x_n y_n}{|x-y|^2}} h^{\frac{\alpha}{2}-1} dh \right) f(y, v(y)) dy \\ &\geq C \int_{\mathbb{R}_+^n \cap \{|y| \leq \frac{1}{2}\}} \frac{1}{|x-y|^{n-\alpha}} \left(\frac{4x_n y_n}{|x-y|^2} \right)^{\frac{\alpha}{2}} f(y, v(y)) dy \\ &\geq C \frac{x_n^{\frac{\alpha}{2}}}{|x|^n} \int_{\mathbb{R}_+^n \cap \{|y| \leq \frac{1}{2}\}} y_n^{\frac{\alpha}{2}} |y|^a v^p(y) dy \\ &\geq C \frac{x_n^{\frac{\alpha}{2}}}{|x|^n} \end{aligned} \quad (3.2)$$

for all $|x| \geq 1$ and $x \in \mathbb{R}_+^n$. Similarly, we can also get that

$$v(x) \geq C \frac{x_n^{\frac{\beta}{2}}}{|x|^n} \int_{\mathbb{R}_+^n \cap \{|y| \leq \frac{1}{2}\}} y_n^{\frac{\alpha}{2}} |y|^b u^q(y) dy \geq C \frac{x_n^{\frac{\beta}{2}}}{|x|^n} \quad (3.3)$$

for all $|x| \geq 1$ and $x \in \mathbb{R}_+^n$.

Next, we will apply the method of scaling spheres to show the following lower bound estimates for the asymptotic behavior of positive solution (u, v) as $|x| \rightarrow +\infty$, which will contradict with the IEs' system (1.10).

Theorem 3.1 Assume $n \geq 1$, $n > \max\{\alpha, \beta\}$, $0 < \alpha, \beta \leq 2$, $f(x, v)$ and $g(x, u)$ satisfy assumptions (A_1) , (A_2) and (A_3) . Suppose (u, v) is a pair of positive solution to IEs' system (1.10), if $f(x, v)$ is subcritical and satisfies (A_3) with $1 \leq p < \frac{n+\alpha+2a}{n-\beta}$ or $g(x, u)$ is subcritical and satisfies (A_3) with $1 \leq q < \frac{n+\beta+2b}{n-\alpha}$, then (u, v) satisfies the following lower bound estimates: for all $x \in \mathbb{R}_+^n$ satisfying $|x| \geq 1$ and $x_n \geq \frac{|x|}{\sqrt{n}}$:

$$u(x) \geq C_\kappa |x|^\kappa, \quad v(x) \geq C_\kappa |x|^\kappa, \quad \forall \kappa < +\infty. \quad (3.4)$$

Proof Given any $\lambda > 0$, we first define the Kelvin transform of functions $u : \mathbb{R}^n \rightarrow \mathbb{R}$ and $v : \mathbb{R}^n \rightarrow \mathbb{R}$ centered at 0 by

$$u_\lambda(x) = \left(\frac{\lambda}{|x|}\right)^{n-\alpha} u\left(\frac{\lambda^2 x}{|x|^2}\right) \quad \text{and} \quad v_\lambda(x) = \left(\frac{\lambda}{|x|}\right)^{n-\beta} v\left(\frac{\lambda^2 x}{|x|^2}\right) \quad (3.5)$$

for arbitrary $x \in \mathbb{R}^n \setminus \{0\}$. It is obvious that the Kelvin transform u_λ, v_λ may have singularity at 0, $\lim_{|x| \rightarrow \infty} |x|^{n-\alpha} u_\lambda(x) = \lambda^{n-\alpha} u(0) = 0$ and $\lim_{|x| \rightarrow \infty} |x|^{n-\beta} v_\lambda(x) = \lambda^{n-\beta} v(0) = 0$. By (3.5), one can infer from the regularity assumptions on (u, v) that $(u_\lambda, v_\lambda) \in C(\overline{\mathbb{R}_+^n} \setminus \{0\}) \times C(\overline{\mathbb{R}_+^n} \setminus \{0\})$.

Next, we will carry out the process of scaling spheres in $\overline{\mathbb{R}_+^n}$ with respect to the origin $0 \in \mathbb{R}^n$. For this purpose, we need some definitions.

Let $\lambda > 0$ be an arbitrary positive real number and let the scaling half sphere be

$$S_\lambda^+ := \left\{x \in \overline{\mathbb{R}_+^n} : |x| = \lambda\right\}. \quad (3.6)$$

We define the reflection of x about the half sphere S_λ^+ by $x^\lambda := \frac{\lambda^2 x}{|x|^2}$ and define

$$B_\lambda^+(0) := B_\lambda(0) \cap \mathbb{R}_+^n, \quad \widetilde{B}_\lambda^+(0) := \{x \in \mathbb{R}_+^n : x^\lambda \in B_\lambda^+(0)\}. \quad (3.7)$$

Let $\omega_u^\lambda(x) := u_\lambda(x) - u(x)$ and $\omega_v^\lambda(x) := v_\lambda(x) - v(x)$ for any $x \in \widetilde{B}_\lambda^+(0) \setminus \{0\}$. By the definition of $u_\lambda, v_\lambda, \omega_u^\lambda$ and ω_v^λ , we have

$$\omega_u^\lambda(x) = -\left(\frac{\lambda}{|x|}\right)^{n-\alpha} \omega_u^\lambda(x^\lambda) = -(\omega_u^\lambda)_\lambda(x), \quad (3.8)$$

$$\omega_v^\lambda(x) = -\left(\frac{\lambda}{|x|}\right)^{n-\beta} \omega_v^\lambda(x^\lambda) = -(\omega_v^\lambda)_\lambda(x) \quad (3.9)$$

for every $x \in \overline{B_\lambda^+(0)} \setminus \{0\}$.

We will first show that for $\lambda > 0$ sufficiently small

$$\omega_u^\lambda(x) \geq 0 \quad \text{and} \quad \omega_v^\lambda(x) \geq 0, \quad \forall x \in B_\lambda^+(0). \quad (3.10)$$

Then, we start dilating the half sphere S_λ^+ from near the origin 0 outward as long as (3.10) holds, until its limiting position $\lambda = +\infty$, and derive lower bound estimates on (u, v) in the cone $\mathcal{C} = \left\{x \in \mathbb{R}_+^n \mid x_n > \frac{|x|}{\sqrt{n}}\right\}$ when $n \geq 2$ and $\mathcal{C} = \mathbb{R}_+$ when $n = 1$.

Consequently, the scaling sphere process can be divided into two steps.

Step 1. Start dilating the half sphere S_λ^+ from near $\lambda = 0$. Define

$$(B_\lambda^+)_u^- := \{x \in B_\lambda^+(0) \mid \omega_u^\lambda(x) < 0\} \quad \text{and} \quad (B_\lambda^+)_v^- := \{x \in B_\lambda^+(0) \mid \omega_v^\lambda(x) < 0\}. \quad (3.11)$$

We will show through contradiction arguments that for $\lambda > 0$ sufficiently small

$$(B_\lambda^+)_u^- = \emptyset \quad \text{and} \quad (B_\lambda^+)_v^- = \emptyset. \quad (3.12)$$

We will need the following lemma, which states some basic properties and estimates of Green's functions $G_\gamma^+(x, y)$.

Lemma 3.2 Assume $G_\gamma^+(x, y)$ be the Green functions given in (1.11). Then, we have

$$G_\gamma^+(x, y) \leq \frac{C}{|x - y|^{n-\gamma}}, \quad \forall x, y \in \mathbb{R}_+^n, \quad (3.13)$$

$$G_\gamma^+(x^\lambda, y) = \left(\frac{|x|}{|y|}\right)^{n-\gamma} G_\gamma^+(x, y^\lambda), \quad \forall x, y \in \mathbb{R}_+^n, \quad (3.14)$$

$$G_\gamma^+(x, y) - \left(\frac{\lambda}{|y|}\right)^{n-\gamma} G_\gamma^+(x, y^\lambda) > 0, \quad \forall x, y \in B_\lambda(0) \cap \mathbb{R}_+^n, \quad (3.15)$$

$$G_\gamma^+(x, y) \geq \frac{C}{|x - y|^{n-\gamma}}, \quad \forall x, y \in \mathbb{R}_+^n, \quad x_n > \frac{|x|}{\sqrt{n}}, \quad y_n > \frac{|y|}{\sqrt{n}}, \quad |x - y| < \frac{|x|}{4}. \quad (3.16)$$

Lemma 3.2 can be proved through straightforward calculations, so we omit the details here.

Since (u, v) satisfies the IEs' system (1.10), we have

$$\begin{aligned}
 u(x) &= \int_{\mathbb{R}_+^n} G_\alpha^+(x, y) f(y, v(y)) dy \\
 &= \int_{B_\lambda^+} G_\alpha^+(x, y) f(y, v(y)) dy + \int_{\mathbb{R}_+^n \setminus B_\lambda} G_\alpha^+(x, y) f(y, v(y)) dy \\
 &= \int_{B_\lambda^+} G_\alpha^+(x, y) f(y, v(y)) dy + \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{2n} G_\alpha^+(x, y^\lambda) f(y^\lambda, v(y^\lambda)) dy,
 \end{aligned} \tag{3.17}$$

$$\begin{aligned}
 v(x) &= \int_{\mathbb{R}_+^n} G_\beta^+(x, y) g(y, u(y)) dy \\
 &= \int_{B_\lambda^+} G_\beta^+(x, y) g(y, u(y)) dy + \int_{\mathbb{R}_+^n \setminus B_\lambda} G_\beta^+(x, y) g(y, u(y)) dy \\
 &= \int_{B_\lambda^+} G_\beta^+(x, y) g(y, u(y)) dy + \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{2n} G_\beta^+(x, y^\lambda) g(y^\lambda, u(y^\lambda)) dy.
 \end{aligned} \tag{3.18}$$

Similarly, from IEs' system (1.10), (3.5) and (3.14) in Lemma 3.2, we obtain

$$\begin{aligned}
 u_\lambda(x) &= \int_{\mathbb{R}_+^n} \left(\frac{\lambda}{|x|} \right)^{n-\alpha} G_\alpha^+(x^\lambda, y) f(y, v(y)) dy \\
 &= \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^\lambda) f(y, v(y)) dy + \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{n+\alpha} G_\alpha^+(x, y) f(y^\lambda, v(y^\lambda)) dy,
 \end{aligned} \tag{3.19}$$

$$\begin{aligned}
 v_\lambda(x) &= \int_{\mathbb{R}_+^n} \left(\frac{\lambda}{|x|} \right)^{n-\beta} G_\beta^+(x^\lambda, y) g(y, u(y)) dy \\
 &= \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{n-\beta} G_\beta^+(x, y^\lambda) g(y, u(y)) dy + \int_{B_\lambda^+} \left(\frac{\lambda}{|y|} \right)^{n+\beta} G_\beta^+(x, y) g(y^\lambda, u(y^\lambda)) dy.
 \end{aligned} \tag{3.20}$$

Then, by (3.17), (3.18), (3.19), and (3.20), we arrive at

$$\begin{aligned}
 \omega_u^\lambda(x) &= u_\lambda(x) - u(x) \\
 &= \int_{B_\lambda^+} \left[G_\alpha^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^\lambda) \right] \left[\left(\frac{\lambda}{|y|} \right)^{n+\alpha} f(y^\lambda, v(y^\lambda)) - f(y, v(y)) \right] dy,
 \end{aligned} \tag{3.21}$$

$$\begin{aligned}
 \omega_v^\lambda(x) &= v_\lambda(x) - v(x) \\
 &= \int_{B_\lambda^+} \left[G_\beta^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\beta} G_\beta^+(x, y^\lambda) \right] \left[\left(\frac{\lambda}{|y|} \right)^{n+\beta} g(y^\lambda, u(y^\lambda)) - g(y, u(y)) \right] dy
 \end{aligned} \tag{3.22}$$

for every $x \in \overline{B_\lambda^+(0)} \setminus \{0\}$.

By (3.21), (3.13), and (3.15) in Lemma 3.2, we have for any $x \in (B_\lambda^+)_u^-$

$$\begin{aligned}
 0 &> \omega_u^\lambda(x) \\
 &= \int_{B_\lambda^+} \left[G_\alpha^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^\lambda) \right] \left[\left(\frac{\lambda}{|y|} \right)^{n+\alpha} f(y^\lambda, v(y^\lambda)) - f(y, v(y)) \right] dy \\
 &> \int_{B_\lambda^+} \left(G_\alpha^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^\lambda) \right) [f(y, v_\lambda(y)) - f(y, v(y))] dy \\
 &\geq \int_{(B_\lambda^+)_v^-} \left(G_\alpha^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^\lambda) \right) [f(y, v_\lambda(y)) - f(y, v(y))] dy \\
 &\geq \int_{(B_\lambda^+)_v^-} G_\alpha^+(x, y) [f(y, v_\lambda(y)) - f(y, v(y))] dy \\
 &\geq C \int_{(B_\lambda^+)_v^-} \frac{1}{|x-y|^{n-\alpha}} \cdot \frac{f(y, v_\lambda(y)) - f(y, v(y))}{v_\lambda(y) - v(y)} \omega_v^\lambda(y) dy,
 \end{aligned} \tag{3.23}$$

where we have used the subcritical condition on $f(x, v)$ for $\mu = \left(\frac{\lambda}{|y|} \right)^{n-\alpha} > 1$ to derive the second inequality and the assumption (A_1) on $f(x, v)$ to derive the third and fourth inequalities. Similarly, by the subcritical or critical condition and assumption (A_1) on $g(x, u)$, (3.22), (3.13) and (3.15) in Lemma 3.2, we have for any $x \in (B_\lambda^+)_v^-$

$$\begin{aligned}
 0 &> \omega_v^\lambda(x) \\
 &= \int_{B_\lambda^+} \left[G_\beta^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\beta} G_\beta^+(x, y^\lambda) \right] \left[\left(\frac{\lambda}{|y|} \right)^{n+\beta} g(y^\lambda, u(y^\lambda)) - g(y, u(y)) \right] dy \\
 &\geq \int_{B_\lambda^+} \left(G_\beta^+(x, y) - \left(\frac{\lambda}{|y|} \right)^{n-\beta} G_\beta^+(x, y^\lambda) \right) [g(y, u_\lambda(y)) - g(y, u(y))] dy \\
 &\geq \int_{(B_\lambda^+)_u^-} G_\beta^+(x, y) [g(y, u_\lambda(y)) - g(y, u(y))] dy \\
 &\geq C \int_{(B_\lambda^+)_u^-} \frac{1}{|x-y|^{n-\beta}} \cdot \frac{g(y, u_\lambda(y)) - g(y, u(y))}{u_\lambda(y) - u(y)} \omega_u^\lambda(y) dy.
 \end{aligned} \tag{3.24}$$

By (3.23), (3.24), Hardy–Littlewood–Sobolev inequality, and Hölder inequality, we obtain for arbitrary $\max \left\{ \frac{n}{n-\alpha}, \frac{n}{n-\beta} \right\} < r < \infty$

$$\begin{aligned}
& \left\| \omega_u^\lambda \right\|_{L^r((B_\lambda^+)_u^-)} \\
& \leq C \left\| \int_{(B_\lambda^+)_v^-} \frac{1}{|x-y|^{n-\alpha}} \cdot \frac{f(y, v_\lambda(y)) - f(y, v(y))}{v_\lambda(y) - v(y)} \omega_v^\lambda(y) dy \right\|_{L^r((B_\lambda^+)_u^-)} \\
& \leq C \left\| \frac{f(x, v_\lambda(x)) - f(x, v(x))}{v_\lambda(x) - v(x)} \omega_v^\lambda(x) \right\|_{L^{\frac{nr}{n+\alpha r}}((B_\lambda^+)_v^-)} \\
& \leq C \left\| \frac{f(x, v_\lambda(x)) - f(x, v(x))}{v_\lambda(x) - v(x)} \right\|_{L^{\frac{n}{\alpha}}((B_\lambda^+)_v^-)} \left\| \omega_v^\lambda \right\|_{L^r((B_\lambda^+)_v^-)},
\end{aligned} \tag{3.25}$$

and

$$\begin{aligned}
& \left\| \omega_v^\lambda \right\|_{L^r((B_\lambda^+)_v^-)} \\
& \leq C \left\| \int_{(B_\lambda^+)_u^-} \frac{1}{|x-y|^{n-\beta}} \cdot \frac{g(y, u_\lambda(y)) - g(y, u(y))}{u_\lambda(y) - u(y)} \omega_u^\lambda(y) dy \right\|_{L^r((B_\lambda^+)_v^-)} \\
& \leq C \left\| \frac{g(x, u_\lambda(x)) - g(x, u(x))}{u_\lambda(x) - u(x)} \omega_u^\lambda(x) \right\|_{L^{\frac{nr}{n+\beta r}}((B_\lambda^+)_u^-)} \\
& \leq C \left\| \frac{g(x, u_\lambda(x)) - g(x, u(x))}{u_\lambda(x) - u(x)} \right\|_{L^{\frac{n}{\beta}}((B_\lambda^+)_u^-)} \left\| \omega_u^\lambda \right\|_{L^r((B_\lambda^+)_u^-)}.
\end{aligned} \tag{3.26}$$

Therefore, we have

$$\begin{aligned}
\left\| \omega_u^\lambda \right\|_{L^r((B_\lambda^+)_u^-)} & \leq C \left\| \frac{f(x, v_\lambda(x)) - f(x, v(x))}{v_\lambda(x) - v(x)} \right\|_{L^{\frac{n}{\alpha}}((B_\lambda^+)_v^-)} \\
& \quad \times \left\| \frac{g(x, u_\lambda(x)) - g(x, u(x))}{u_\lambda(x) - u(x)} \right\|_{L^{\frac{n}{\beta}}((B_\lambda^+)_u^-)} \left\| \omega_u^\lambda \right\|_{L^r((B_\lambda^+)_u^-)}.
\end{aligned} \tag{3.27}$$

Since $(u, v) \in C(\overline{\mathbb{R}^n}) \times C(\overline{\mathbb{R}^n})$, $f(x, v)$ and $g(x, u)$ satisfy the assumption $(\mathbf{A}_2)$, there exists a $\delta_0 > 0$ sufficiently small, such that

$$C \left\| \frac{f(x, v_\lambda(x)) - f(x, v(x))}{v_\lambda(x) - v(x)} \right\|_{L^{\frac{n}{\alpha}}((B_\lambda^+)_v^-)} \left\| \frac{g(x, u_\lambda(x)) - g(x, u(x))}{u_\lambda(x) - u(x)} \right\|_{L^{\frac{n}{\beta}}((B_\lambda^+)_u^-)} < \frac{1}{2} \tag{3.28}$$

for any $0 < \lambda < \delta_0$. Thus, by (3.26) and (3.27), we must have for any $0 < \lambda < \delta_0$

$$\left\| \omega_u^\lambda \right\|_{L^r((B_\lambda^+)_u^-)} = 0, \quad \left\| \omega_v^\lambda \right\|_{L^r((B_\lambda^+)_v^-)} = 0. \tag{3.29}$$

Note that $(u_\lambda, v_\lambda) \in C(\overline{\mathbb{R}_+^n} \setminus \{0\}) \times C(\overline{\mathbb{R}_+^n} \setminus \{0\})$, and we immediately get that $(B_\lambda^+)_u^- = (B_\lambda^+)_v^- = \emptyset$. Therefore, (3.10) holds true for any $0 < \lambda < \delta_0$. This completes Step 1.

Step 2. Dilate the half sphere S_λ^+ outward until $\lambda = +\infty$ to derive lower bound estimates on u in a cone. Step 1 provides us a start point to dilate the half sphere S_λ^+ from near $\lambda = 0$. Now, we dilate the half sphere S_λ^+ outward as long as (3.10) holds. Let

$$\lambda_0 := \sup \left\{ \lambda > 0 \mid \omega_u^\mu \geq 0 \text{ and } \omega_v^\mu \geq 0 \text{ in } B_\mu^+(0), \forall 0 < \mu \leq \lambda \right\} \in (0, +\infty], \quad (3.30)$$

and hence, one has

$$\omega_u^{\lambda_0}(x) \geq 0 \quad \text{and} \quad \omega_v^{\lambda_0}(x) \geq 0, \quad \forall x \in B_{\lambda_0}^+(0). \quad (3.31)$$

In what follows, we will prove $\lambda_0 = +\infty$ by contradiction arguments. Suppose on contrary that $0 < \lambda_0 < +\infty$. To get a contradiction, we will first prove

$$\omega_u^{\lambda_0}(x) > 0 \quad \text{and} \quad \omega_v^{\lambda_0}(x) > 0, \quad \forall x \in B_{\lambda_0}^+(0). \quad (3.32)$$

Indeed, since $f(x, v)$ is subcritical and satisfies the assumption $(\mathbf{A}_1)$, one can derive from (3.15) in Lemma 3.2, (3.21) and (3.31) that for any $x \in B_{\lambda_0}^+(0)$

$$\begin{aligned} \omega_u^{\lambda_0}(x) &= u_{\lambda_0}(x) - u(x) \\ &= \int_{B_{\lambda_0}^+} \left[G_\alpha^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^{\lambda_0}) \right] \left[\left(\frac{\lambda_0}{|y|} \right)^{n+\alpha} f(y^{\lambda_0}, v(y^{\lambda_0})) - f(y, v(y)) \right] dy \\ &> \int_{B_{\lambda_0}^+} \left(G_\alpha^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\alpha} G_\alpha^+(x, y^{\lambda_0}) \right) [f(y, v_{\lambda_0}(y)) - f(y, v(y))] dy \geq 0. \end{aligned} \quad (3.33)$$

If $g(x, u)$ is subcritical, one can derive from the assumption $(\mathbf{A}_1)$, (3.15) in Lemma 3.2, (3.22) and (3.31) that for any $x \in B_{\lambda_0}^+(0)$

$$\begin{aligned} \omega_v^{\lambda_0}(x) &= v_{\lambda_0}(x) - v(x) \\ &= \int_{B_{\lambda_0}^+} \left[G_\beta^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\beta} G_\beta^+(x, y^{\lambda_0}) \right] \left[\left(\frac{\lambda_0}{|y|} \right)^{n+\beta} g(y^{\lambda_0}, u(y^{\lambda_0})) - g(y, u(y)) \right] dy \\ &> \int_{B_{\lambda_0}^+} \left(G_\beta^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\beta} G_\beta^+(x, y^{\lambda_0}) \right) [g(y, u_{\lambda_0}(y)) - g(y, u(y))] dy \geq 0. \end{aligned} \quad (3.34)$$

If $g(x, u)$ has critical growth, one can derive from the assumption $(\mathbf{A}_1)$, (3.15) in Lemma 3.2, (3.22) and (3.33) that for any $x \in B_{\lambda_0}^+(0)$

$$\begin{aligned}
 \omega_v^{\lambda_0}(x) &= v_{\lambda_0}(x) - v(x) \\
 &= \int_{B_{\lambda_0}^+} \left[G_{\beta}^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\beta} G_{\beta}^+(x, y^{\lambda_0}) \right] \left[\left(\frac{\lambda_0}{|y|} \right)^{n+\beta} g(y^{\lambda_0}, u(y^{\lambda_0})) - g(y, u(y)) \right] dy \\
 &= \int_{B_{\lambda_0}^+} \left(G_{\beta}^+(x, y) - \left(\frac{\lambda_0}{|y|} \right)^{n-\beta} G_{\beta}^+(x, y^{\lambda_0}) \right) [g(y, u_{\lambda_0}(y)) - g(y, u(y))] dy > 0.
 \end{aligned} \tag{3.35}$$

Thus, we have arrived at (3.32).

Next, we will obtain a contradiction with (3.30) by showing that the half sphere S_{λ}^+ can be dilated outward a little bit further; more precisely, there exists an $\varepsilon > 0$ small enough, such that $\omega_u^{\lambda} \geq 0$ and $\omega_v^{\lambda} \geq 0$ in $B_{\lambda}^+(0)$ for all $\lambda \in [\lambda_0, \lambda_0 + \varepsilon]$.

To this end, let $\delta_1 > 0$ be sufficiently small, which will be determined later. Define the narrow region

$$A_{\delta_1} := \left\{ x \in B_{\lambda_0}^+(0) \mid \text{dist}\left(x, \partial(B_{\lambda_0}^+(0))\right) < \delta_1 \right\}, \tag{3.36}$$

that is, $A_{\delta_1} = \left\{ x \in B_{\lambda_0}^+(0) \mid x_n < \delta_1 \text{ or } \lambda_0 - \delta_1 < |x| < \lambda_0 \right\}$.

Since $\omega_u^{\lambda_0}$ and $\omega_v^{\lambda_0}$ are continuous in $\overline{\mathbb{R}_+^n} \setminus \{0\}$ and $A_{\delta_1}^c := (B_{\lambda_0}^+(0)) \setminus A_{\delta_1}$ is a compact subset, Eq. (3.32) implies that there exists a positive constant $m_0 > 0$, such that

$$\omega_u^{\lambda_0}(x) > m_0 \quad \text{and} \quad \omega_v^{\lambda_0}(x) > m_0, \quad \forall x \in A_{\delta_1}^c. \tag{3.37}$$

By continuity, we can choose $\delta_2 > 0$ (depending on δ_1) sufficiently small, such that for any $\lambda \in [\lambda_0, \lambda_0 + \delta_2]$

$$\omega_u^{\lambda}(x) > \frac{m_0}{2} \quad \text{and} \quad \omega_v^{\lambda}(x) > \frac{m_0}{2}, \quad \forall x \in A_{\delta_1}^c. \tag{3.38}$$

Hence, $(B_{\lambda}^+)_u^-$ and $(B_{\lambda}^+)_v^-$ must be contained in the narrow region $A_{\delta_1} \cup ((B_{\lambda} \setminus B_{\lambda_0}) \cap \mathbb{R}_+^n)$. Consequently, we can choose δ_1 sufficiently small (and δ_2 more smaller if necessary), such that for any $\lambda \in [\lambda_0, \lambda_0 + \delta_2]$

$$C \left\| \frac{f(x, v_{\lambda}(x)) - f(x, v(x))}{v_{\lambda}(x) - v(x)} \right\|_{L^{\frac{n}{\alpha}}((B_{\lambda}^+)_v^-)} \left\| \frac{g(x, u_{\lambda}(x)) - g(x, u(x))}{u_{\lambda}(x) - u(x)} \right\|_{L^{\frac{n}{\beta}}((B_{\lambda}^+)_u^-)} < \frac{1}{2}, \tag{3.39}$$

where C is the constant in the last inequality of (3.27). Then, by (3.26), (3.27) and the same arguments as in step 1, we obtain that for any $\lambda \in [\lambda_0, \lambda_0 + \delta_2]$

$$\|\omega_u^{\lambda}\|_{L^r((B_{\lambda}^+)_u^-)} = \|\omega_v^{\lambda}\|_{L^r((B_{\lambda}^+)_v^-)} = 0, \quad \forall \max \left\{ \frac{n}{n-\alpha}, \frac{n}{n-\beta} \right\} < r < \infty, \tag{3.40}$$

and hence

$$\omega_u^{\lambda}(x) \geq 0 \quad \text{and} \quad \omega_v^{\lambda}(x) \geq 0, \quad \forall x \in B_{\lambda}^+(0). \tag{3.41}$$

This contradicts the definition of λ_0 . Thus, we must have $\lambda_0 = +\infty$. Then, it follows immediately that:

$$\begin{cases} u(x) \geq \left(\frac{\lambda}{|x|}\right)^{n-\alpha} u\left(\frac{\lambda^2 x}{|x|^2}\right), & \forall |x| \geq \lambda, \quad \forall 0 < \lambda < +\infty; \\ v(x) \geq \left(\frac{\lambda}{|x|}\right)^{n-\beta} v\left(\frac{\lambda^2 x}{|x|^2}\right), & \forall |x| \geq \lambda, \quad \forall 0 < \lambda < +\infty. \end{cases} \quad (3.42)$$

For arbitrary $|x| \geq 1$, $x \in \mathbb{R}_+^n$, let $\lambda := \sqrt{|x|}$, and then, (3.42) yields that

$$\begin{cases} u(x) \geq \frac{1}{|x|^{\frac{n-\alpha}{2}}} u\left(\frac{x}{|x|}\right). \\ v(x) \geq \frac{1}{|x|^{\frac{n-\beta}{2}}} v\left(\frac{x}{|x|}\right). \end{cases} \quad (3.43)$$

As a consequence, we arrive at the following lower bound estimates: there exists a $C_0 > 0$, such that

$$\begin{cases} u(x) \geq \left(\min_{|x|=1, x_n \geq \frac{1}{\sqrt{n}}} u(x)\right) \frac{1}{|x|^{\frac{n-\alpha}{2}}} \geq \frac{C_0}{|x|^{\frac{n-\alpha}{2}}}, & \forall |x| \geq 1, \quad x_n \geq \frac{|x|}{\sqrt{n}}; \\ v(x) \geq \left(\min_{|x|=1, x_n \geq \frac{1}{\sqrt{n}}} v(x)\right) \frac{1}{|x|^{\frac{n-\beta}{2}}} \geq \frac{C_0}{|x|^{\frac{n-\beta}{2}}}, & \forall |x| \geq 1, \quad x_n \geq \frac{|x|}{\sqrt{n}}. \end{cases} \quad (3.44)$$

The lower bound estimate (3.44) can be improved remarkably via using the “Bootstrap” iteration technique and the IEs’ system (1.10).

In fact, let $\mu_{u,0} := \frac{n-\alpha}{2}$ and $\mu_{v,0} := \frac{n-\beta}{2}$, and we can infer from the assumption (A₃) on $f(x, v)$ and $g(x, u)$, IEs’ system (1.10), (3.16) in Lemmas 3.2 and (3.44) that for any x satisfying $|x| \geq 1$ and $x_n \geq \frac{|x|}{\sqrt{n}}$

$$\begin{aligned} u(x) &= \int_{\mathbb{R}_+^n} G_\alpha^+(x, y) f(y, v(y)) dy \\ &\geq C \int_{|x-y| < \frac{|x|}{4}, y_n \geq \frac{|y|}{\sqrt{n}}} G_\alpha^+(x, y) \frac{1}{|y|^{p\mu_{v,0}-a}} dy \\ &\geq C \int_{|x-y| < \frac{|x|}{4}, y_n \geq \frac{|y|}{\sqrt{n}}} \frac{1}{|x-y|^{n-\alpha}} \cdot \frac{1}{|y|^{p\mu_{v,0}-a}} dy \\ &\geq \frac{C}{|x|^{p\mu_{v,0}-a}} \int_{|x-y| < \frac{|x|}{4}, y_n \geq \frac{|y|}{\sqrt{n}}} \frac{1}{|x-y|^{n-\alpha}} dy \\ &\geq \frac{C_1}{|x|^{p\mu_{v,0}-(a+\alpha)}}, \end{aligned} \quad (3.45)$$

where we have used the fact that if x is in the cone $\mathcal{C} = \left\{x \in \mathbb{R}^n \mid x_n \geq \frac{|x|}{\sqrt{n}}\right\}$ ($\mathcal{C} = \mathbb{R}_+$ when $n = 1$), then a certain proportion (depending only on n) of the ball centered at x is also contained in this cone $\mathcal{C}$. Let $\mu_{u,1} := p\mu_{v,0} - (a + \alpha)$, and then, (3.45) gives us a better lower bound on u

$$u(x) \geq \frac{C_1}{|x|^{\mu_{u,1}}}, \quad \forall |x| \geq 1, x_n \geq \frac{|x|}{\sqrt{n}}. \quad (3.46)$$

Similarly, let $\mu_{v,1} = q\mu_{u,0} - (b + \beta)$, and then, we also have a better lower bound on v (choose C_1 smaller if necessary)

$$v(x) \geq \frac{C_1}{|x|^{\mu_{v,1}}}, \quad \forall |x| \geq 1, x_n \geq \frac{|x|}{\sqrt{n}}. \quad (3.47)$$

For $k = 0, 1, 2, \dots$, define

$$\begin{aligned} \mu_{u,k+1} &:= p\mu_{v,k} - (a + \alpha), \\ \mu_{v,k+1} &:= q\mu_{u,k} - (b + \beta). \end{aligned} \quad (3.48)$$

Then, by continuing the above iteration process involving the IEs system (1.10), we have

$$u(x) \geq \frac{C_k}{|x|^{\mu_{u,k}}}, \quad v(x) \geq \frac{C_k}{|x|^{\mu_{v,k}}}, \quad \forall |x| \geq 1, x_n \geq \frac{|x|}{\sqrt{n}}. \quad (3.49)$$

Note that $1 \leq p < \frac{n+\alpha+2a}{n-\beta}$ and $1 \leq q \leq \frac{n+\beta+2b}{n-\alpha}$, and

$$\begin{aligned} \mu_{u,k+2} &:= pq\mu_{u,k} - p(b + \beta) - (a + \alpha), \\ \mu_{v,k+2} &:= pq\mu_{v,k} - q(a + \alpha) - (b + \beta); \end{aligned} \quad (3.50)$$

one can verify that $\mu_{u,2j} \rightarrow -\infty$, $\mu_{v,2j} \rightarrow -\infty$ as $j \rightarrow +\infty$. Consequently, we have proved Theorem 3.1. $\square$

For arbitrarily fixed $x \in \mathbb{R}_+^n$, by (2.14), there exists a $R_x > 0$ sufficiently large, such that for any $|y| \geq R_x$

$$G_\gamma^+(x, y) \geq C \frac{(x_n y_n)^{\frac{\gamma}{2}}}{|y|^n}, \quad (3.51)$$

where $0 < \gamma \leq 2$. Therefore, the IEs' system (1.10) and assumption (A_3) on $f(x, v)$ and $g(x, u)$ imply that

$$\begin{cases} C(x_n)^{\frac{\alpha}{2}} \int_{\mathbb{R}_+^n \cap \mathcal{C} \cap \{|y| \geq R_x\}} \frac{(y_n)^{\frac{\alpha}{2}}}{|y|^{n-a}} v^p(y) dy \leq u(x) < \infty, \\ C(x_n)^{\frac{\beta}{2}} \int_{\mathbb{R}_+^n \cap \mathcal{C} \cap \{|y| \geq R_x\}} \frac{(y_n)^{\frac{\beta}{2}}}{|y|^{n-b}} u^q(y) dy \leq v(x) < \infty. \end{cases} \quad (3.52)$$

It is evident that the lower bounds in Theorem 3.1 contradict the integrability of (u, v) in (3.52). Thus, we must have $(u, v) \equiv (0, 0)$ in $\mathbb{R}_+^n$.

This concludes our proof of Theorem 1.5.

Acknowledgements The authors are grateful to the anonymous referees for their careful reading and valuable comments and suggestions that improved the presentation of the paper.

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